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DISTANCE PERCEPTION IN THE SPINY MOUSE *ACOMYS CAHIRINUS*: VERTICAL JUMPING[1]

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Summary.—*Acomys cahirinus*, a precocial muroid, that has shown precise jumping in the natural habitat, did not jump from 25 cm in a laboratory situation. To investigate this further, *A. cahirinus* were observed jumping from platforms at two different heights, onto different sized checkered substrates and from a visual cliff. Adult animals discriminated between platforms that were 6.4 cm and 25.4 cm above the substrate and between small and large checkered patterns on the floor. Most adult animals and neonates jumped down on the shallow side of the visual cliff. Animals developed individual patterns of jumping over a series of trials, with some jumping often, some rarely, and others jumping only from the low platform. Good distance perception was indicated when they did not jump from heights, and by their making appropriate postural adjustment when they did jump from heights and landed without mishap. Different spacing of trials indicated that height was a more effective stimulus for animals which had all four conditions on the same day, while floor pattern was more effective for animals with each of the four conditions on a separate day.

Acomys cahirinus (spiny mouse) is a desert mouse that lives among rocky ledges in the Middle East. These mice carry their food, that consists primarily of snails and insects, as well as some vegetation, back to their home areas that are usually already existing crevices among rocks (Shkolnik, 1966). In response to predators they run for cover. They pounce upon prey, but do not usually jump for other food, and jumping is not their first response when attacked. However, they have been seen to jump from heights of approximately 1 m vertically and horizontally among rocks (Tobach, personal observation, 1976).

In our laboratory, the adult mice have been observed to jump both horizontally and vertically distances of 30 to 35 cm in both 30-cm cubed cages and in enclosures measuring 100 × 100 × 90 cm. In contrast to these observations, adult *A. cahirinus* have been reported to demonstrate distance discrimination when they jumped primarily to the shallow side of a traditional visual cliff (Routtenberg & Glickman, 1964). Greenberg (1986) also noted they would jump from a Plexiglas platform 5 cm from the substrate, but not from a Plexiglas platform 25 cm above the substrate. These findings seem to contradict our observations in which we saw that they were capable of jump-

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ing down from great distances both in their natural habitat in the desert and in their home cages in the laboratory.

Field and laboratory studies can be complementary in elucidating behavioral processes. When carefully designed laboratory experiments yield results that appear to contradict the observations from the field or from serendipitous daily observations of an animal's behavior in captivity, such discrepancies are usually attributable to failure to present a situation to the animal that is most compatible with its experience as an individual and with the evolutionary history of the animal. The fact that our spiny mice showed behavior in captivity (the colony cages) similar to that shown in their natural habitat indicated that the type of situation given to the spiny mice in the studies cited needed to be examined further. Accordingly, we embarked on a series of experiments designed to help us understand the discrepancy. We report here on the first set of such experiments.

We presented the animals with different platform heights and a visual cliff to replicate the studies of Greenberg and of Routtenberg and Glickman with our population of animals. We extended their procedures by using additional substrates and observing the animals over many trials to increase their experience with the experimental situation. The use of floors with large and small checkers allowed the animals to make more discriminations than in previous experiments. We also looked at the phase of the light/dark cycle and the spacing of trials, two variables which have been important in other experimental situations.

Experiment 1

Our first set of experiments consisted of a modification of the checkered substrate used by Greenberg (1986) and other substrates that varied the size of the checkers. Trials continued over a 4-week period, instead of the single trial usually used in this type of experiment, giving the animals the opportunity to respond to each stimulus configuration four times.

Method

Animals.—Eight male and eight female *A. cahirinus*, one to three years in age, were housed individually in metal cages ($30.5 \times 35.4 \times 30.5$ cm) with sand on the solid floor pan. Three walls were solid metal; the front and top of the cage were wire mesh (1.27 cm square). Laboratory rodent chow and carrots or mealworms were given daily to the animals after observation. The rooms were maintained at 20°C on a 12-hr. dark/light cycle. The animals had not been in any prior experiments.

Apparatus.—The apparatus consisted of a circular metal enclosure, 80 cm in diameter, 61 cm high, painted gray. The enclosure was placed on top of one of four different wooden floors, each covered by a Plexiglas sheet. The floors were (1) painted the same gray as the walls, (2) covered by black and white checkers 2.5 cm in size (large) and made from glossy photographs, (3) black and white checkers 0.6-cm square (small) from glossy photographs, (4) half covered by the large checkers and half by the small. Placed in the center of the floor was a 30-cm diameter Plexiglas disc on which one of two metal shafts of different heights, 25.4 or 6.4 cm, could hold another 25.4-cm Plexiglas disc as a platform. The enclosure was illuminated by a single lamp, suspended from the ceiling, which produced approximately 43 lx of incandescent light (Lunasix light meter) inside the enclosure so that no shadows were cast.

Procedure.—Each animal was captured in an opaque plastic cup in its home cage, carried to a separate observation room (within approximately 1 min.), and placed in the center of the Plexiglas platform. The cup was removed, thus starting a 2-min. trial. An animal that did not jump was removed by cup after 2 min. and returned to its home cage. Timing was done with stopwatches. All Plexiglas surfaces were cleaned after each observation with 70% alcohol and water.

Eight female and eight male *A. cahirinus* were all observed for one trial in the morning and one in the afternoon during the light phase of the light/dark cycle. Each floor pattern was used one day a week for four weeks (Tuesday through Friday). All sequences including the two platform heights were assigned according to a Latin Square. If the animal jumped from the platform, it was caught with the cup immediately for the first three weeks. During the last week it was allowed to stay inside the enclosure until the 2 min. had elapsed. Analysis of data indicated that the differences found were not related to this change in procedure. An observer sat in the same location for every trial and recorded the latency and the area to which the animal jumped. All statistics used for data analyses were nonparametric (Siegel, 1956).

Results and Discussion

The animals demonstrated good discrimination of platform height regardless of substrate (Tables 1 and 2); significantly fewer animals jumped from 25.4 cm than from 6.4 cm (Cochran Q, $p < 0.001$). This agrees with the previous study (Greenberg, 1986). The differences between the number of animals jumping from the two heights were not related to the sequences of stimulus presentation or animal observation or to the age or sex of the animals.

The modification of the procedures brought out other aspects of the jumping behavior. The behavior of the animals on the first day's trial was not indicative of their over-all jumping pattern. Many animals developed individual patterns of responding over the course of the experiment. As the experiment continued into the third and fourth weeks over-all jumping declined, but while six animals stopped jumping from both platforms three

TABLE 1
TOTAL NUMBER OF *Acomys cahirinus* JUMPING ($N = 16$) AND THEIR MEDIAN LATENCIES (SEC.) FROM THE 25.4-CM PLATFORM

Pattern on Floor		Presentations			
		First	Second	Third	Fourth
Large Checkers	Number	4	6	5	5
	Latency				
	Median	70.0	63.0	57.0	21.0
	Range	44–102	24–110	2–113	5–77
Small Checkers	Number	3	2	2	6
	Latency				
	Median	70.0	59.0	61.5	81.0
	Range	24–115	13–105	60–63	16–115
Gray	Number	2	2	1	1
	Latency				
	Median	56.0	93.5	25.0	91.0
	Range	23–89	70–117		
Large/Small Checkers	Jumped to Large				
	Number	0	2	4	3
	Latency				
	Median		57.0	79.5	79.0
	Range		40–74	31–118	29–119
	Jumped to Small				
	Number	1	0	1	0
	Latency				
	Median	47.0		15.0	
	Range				

others began jumping more frequently. Seven animals (nonjumpers) made significantly fewer jumps, out of a possible 32 jumps during the entire experiment, than expected by chance (Binomial, $p = 0.05$) and two others (jumpers) made significantly more jumps than chance. The other seven were in between. Twelve out of the 16 animals jumped significantly less from the high platform than the low one regardless of substrate (Binomial, $p = 0.05$). The other four did jump less from the high platform, though not significantly, and three of these animals had the highest over-all numbers of jumps.

Although none of Greenberg's *A. cahirinus* jumped from the high platform during the two opportunities they had, 13 of our animals jumped at least once. These animals demonstrated that they are capable of making the postural adjustments necessary to jump successfully from the 25.4-cm platform: six animals jumped once, four animals jumped three to five times, and the three animals that jumped most during the entire experiment jumped from the high platform 7, 9, and 13 times, out of a possible 16.

Table 1 shows that most of the animals did not jump from the 25.4-cm platform. The number of animals that jumped, as well as their latencies, were not related to the type of floor pattern or to the presentation sequence. Because the number of animals that jumped was small and the individual jumping patterns were variable, the data for the entire population of 16 was included in the analysis of latencies. There were no significant differences in the latencies analyzed by floor pattern (Friedman analysis of variance).

Table 2 shows that more animals jumped with shorter latencies from the 6.4-cm platform than from the 25.4-cm platform. By inspection, the latencies for those that jumped were not related to floor pattern. Again, the number of animals that jumped was variable and the latency data analyses included the entire population of 16. These data, which are presented in Table 3, do show latency differences related to floor pattern. Most of the animals did jump from the 6.4-cm height when the floor was gray or had large checkers on it on the first presentation. On the first presentation of the small checkers, fewer animals jumped, resulting in significantly longer latencies than those to the other three floors (Friedman $\chi_r^2 = 18.6$, $p < 0.001$).

The 6.4-cm height of the lower platform was chosen so that the small checkers when viewed from this height would have approximately the same

TABLE 2
TOTAL NUMBER OF *Acomys cahirinus* JUMPING ($N = 16$) AND THEIR MEDIAN LATENCIES (SEC.) FROM THE 6.4-CM PLATFORM

Pattern on Floor		Presentations			
		First	Second	Third	Fourth
Large Checkers	Number	15	13	11	11
	Latency				
	Median	8.0	9.0	5.0	20.0
	Range	4–89	2–89	3–103	4–87
Small Checkers	Number	7	11	10	9
	Latency				
	Median	17.0	20.0	26.0	4.0
	Range	4–93	3–96	5–96	2–68
Gray	Number	14	10	6	8
	Latency				
	Median	18.0	16.0	6.0	12.0
	Range	5–92	3–85	3–50	2–100
Large/Small Checkers	Jumped to Large				
	Number	13	11	9	10
	Latency				
	Median	7.0	8.0	19.0	19.0
	Range	1–70	3–74	6–83	3–107
	Jumped to Small				
	Number	0	3	3	1
	Latency				
	Median		9.0	21.0	21.0
	Range		7–50	15–119	

TABLE 3
TOTAL NUMBER OF *Acomys cahirinus* JUMPING ($N = 16$) AND THE MEDIAN LATENCIES (SEC.), INCLUDING NONJUMPERS, FROM THE 6.4-CM PLATFORM

Pattern on Floor		Presentations			
		First*	Second	Third	Fourth
Large Checkers	Number	15	13	11	11
	Latency				
	Median	8.5	15.5	22.0	36.5
	Range	4–120	2–120	3–120	4–120
Small Checkers	Number	7	11	10	9
	Latency†				
	Median	120.0	48.0	42.0	66.5
	Range	4–120	3–120	5–120	2–120
Gray	Number	14	10	6	8
	Latency†				
	Median	20.0	50.0	120.0	110.0
	Range	5–120	3–120	3–120	2–120
Split Floor	Number	13	14	12	11
Large/Small	Latency				
Checkers	Median	10.5	9.0	44.0	59.5
	Range	1–120	3–120	6–120	3–120

*Friedman analysis of variance, $p < 0.001$ for latency comparison among floor patterns on first presentation only.
†Wilcoxon test, $p < 0.05$ for comparison of first and second presentation of 6.4-cm platform with small checkers and gray floor only.

visual angle as the large checkers when seen from the 25.4-cm height. *A. cahirinus* demonstrated that they were using cues from the floor patterns when they hesitated to jump from the 6.4-cm platform to the small checkered floor. When the floor was half large and half small checkers, the animals jumped to the large checkers more frequently from both platforms (Tables 1 and 2). Ten animals did not jump to small at all when presented with the large checkers, although eight of these animals did jump to small when alone.

Latencies were significantly reduced from the first to second presentation of the small squares (Wilcoxon, $z = 1.96$, $p < 0.05$), but significantly increased from the first to second presentation of the gray floor (Wilcoxon, $z = 1.97$, $p < 0.05$). The latencies for jumping from the lower platform to all floor patterns, except the small checkers, increased with repeated presentations, although the change over days was not statistically significant. Walk and Gibson (1961) also found longer latencies and fewer descents by their rats over repeated trials on a visual cliff with a 1.9-cm gray and white checkered pattern.

EXPERIMENT 2

To verify that our population would behave similarly to the spiny mice observed by Routtenberg and Glickman (1964) on a traditional visual cliff, we observed the animals that had been in the platform experiment on a visual cliff.

Method

Animals.—These were the animals in Exp. 1. The observations took place after Exp. 1 was completed.

Apparatus.—A 60-cm square metal enclosure 53 cm high was placed on a sheet of Plexiglas on top of a second 60-cm square metal enclosure. A 2.5-cm black and white checkered pattern was placed flush against the underside of the Plexiglas in one half of the enclosure and on the floor 53 cm below the Plexiglas in the other half. A 7- × 60-cm wooden block, 5 cm high, also covered with checkers was placed on top of the Plexiglas across the intersection of the two halves. The lighting was the same as in Exp. 1.

Procedure.—Each animal was captured in an opaque plastic cup in its home cage, carried to a separate observation room, and placed in the center of the checkered wooden block. A 2-min. trial was started with the removal of the cup. All animals that stepped off the block were removed by cup immediately, those that remained on the block were removed after 2 min.; all 16 animals were observed in one morning.

In the afternoon, the wooden block was removed from the visual cliff and the 25.4-cm platform was placed in the center of the square enclosure so that it was equally across both the shallow and deep sides. All 16 animals were observed in the afternoon on the 25.4-cm platform with the same procedure as was described in Exp. 1 in which the animals that jumped were captured immediately.

Results and Discussion

Eleven of the 16 animals stepped down onto the shallow side; the other five remained on the block. Of the 11, eight had been most likely to jump from the 6.4-cm platform in Exp. 1, six of these had made most of the jumps from the 25.4-cm platform. Four of the animals that remained on the center block of the visual cliff were nonjumpers in Exp. 1. When the 25.4-cm transparent platform was placed so that half was over the shallow side and half over the deep side only six animals jumped, all to the shallow side. All of these animals had stepped off the visual cliff center block; four were the animals with the most jumps in Exp. 1 and one was a nonjumper.

Fewer of our animals jumped from the visual cliff center block than Routtenberg and Glickman's animals, 69% vs 85%, respectively, but our animals had been observed for the previous 4 weeks and the percentage stepping onto the shallow side of the visual cliff is comparable to the results from the 6.4-cm platform to the large checkers in the third and fourth weeks. The response latencies for the animals that jumped down on the traditional visual cliff were also comparable and ranged from 35 to 70 sec., with a median latency of 25 sec. The response latencies for those that jumped from the 25.4-cm platform on top of the visual cliff ranged from 29 sec. to 115 sec., with a median of 52 sec. This result is similar to the jumping latencies observed in Exp. 1 with the 25.4-cm platform on the large checkered pattern.

Experiment 3

Since the adult spiny mice responded to the visual cliff situation as would be expected from observations with other adult mammals, it was important to look at neonates, as spiny mice are precocial at birth. *A. cahirinus* have a long gestation period compared with other rodents. They are born furred and shortly after birth their eyes and ears are open, and they walk around.

Method

Animals.—Twenty-two neonates were observed in the apparatus on their day of birth. The pregnant females were housed in a dark room so the neonates did not receive any light stimulation until they were brought to the observation room. All animals' eyes were open. Four additional neonates were observed about two days after birth since they were born on the weekend. The size of the 13 litters ranged from one to four pups; the modal size was two.

Apparatus.—A wooden box, painted with dark gray latex paint and polyurethane, had two 15-cm square platforms, similarly painted, which were separated by a barrier 0.6-cm wide on which the animal was placed. One platform was 4 cm below the barrier and one was 25 cm below. The walls of the box extended 25 cm above the top of the barrier.

Procedure.—Each animal was cupped in its home cage and carried to the lighted observation room. To place the animal on the narrow center barrier it was necessary to pick it up by hand. Animals that jumped or fell were removed immediately. Those remaining on the barrier were removed after 2 min. had elapsed.

Results and Discussion

Sixteen of the animals observed on their days of birth jumped down 4 cm to the shallow side. This result is significant (Binomial, $p = 0.004$). Of these, 14 jumped within the first 30 sec., one jumped between 30 and 60 sec., and the other jumped between 90 and 120 sec. Two jumped to the deep side (25 cm) within the first 30 sec., two others fell into the deep side at 20 and 60 sec., and the last two remained on the barrier. One of these last animals did jump to the shallow side after the trial was over. Thus, most of these animals responded in the manner usually interpreted as demonstrating depth perception within hours of their birth.

However, the four other animals born during the weekend and observed about two days after birth would not stay on the barrier. Two jumped to the deep side and climbed back up the wooden wall. A third spiny mouse continued to climb up the wall from the barrier out of the apparatus. The fourth, while on the barrier, ran to the wall perpendicular to the barrier but fell into the shallow side. These animals did not show the typical caution when presented with a choice between a shallow and a deep area; rather they showed good visual-motor coordination in response to a complex environment with different depth characteristics.

Experiment 4

Exp. 1 had been complicated by the four different floors, each one presented on a different day. To reduce the number of substrate choices, we used only the large or small checkered floors. To investigate the effect of daily stimulus presentations, different groups of younger animals were either given four stimulus combinations on the same day or on four consecutive days.

Method

Animals.—Sixteen male and eight female *A. cahirinus*, 1 to 5 months in age, were maintained as described for Exp. 1 except that the light/dark cycle was reversed and water bottles were made available for 2 hr. every weekday morning. Ten animals had been placed on a visual cliff shortly after birth (Exp. 3); the other 14 had not been in any prior experiments. These animals had been reared in large enclosures (100 × 100 × 90 cm) until weaning. The enclosures had sheet metal walls which were painted sand color on the bottom half and sky blue on the top, and sand and rocks on the substrate.

Apparatus.—The apparatus was the same as that used in Exp. 1, but only the large and small checkered floors were presented.

Procedure.—The procedure was the same as that of Exp. 1. All animals that jumped were removed by the cup from the apparatus immediately, and those that did not jump were removed after 2 min. Different stimulus combinations consisted of either large or small checkered floor with either the 6.4- or 25.4-cm platform. Twelve animals (Group A), eight males and four females, received all four combinations sequentially on the same day. The animal was placed into an empty glass tank in between trials. During the following week, 12 other animals (Group B) had one trial a day with a different stimulus combination on four consecutive days. All sequences were counterbalanced for groups of three animals.

Results and Discussion

No differences were related to age, sex, sequence, or experimental experience. Table 4 shows the total number of animals jumping and the median latencies and ranges for the two groups; A, when all four stimulus combinations were presented in one session and B, when the four stimulus combinations were presented on separate days. When the effect of height was investigated, significantly fewer animals in Group A jumped from the 25.4-cm platform than from the lower platform to either size of checkers (McNemar Test of Change, $p = 0.019$ for the small checkers, and $p = 0.016$ for the large, one-tailed test). Similar jumping patterns were found for the small checkers for Group B when the high and low platforms were compared (McNemar Test of Change, $p = 0.031$), but for the large checkers for Group B, the effect of platform height was not significant. With regard to the number of animals jumping to all stimulus combinations, there were no significant differences between Groups A and B.

Analyses of the jump latencies also show differences. When the stimulus combinations were presented on the same day, Group A, there were significantly longer latencies from the 25.4-cm platform than from the 6.4-

TABLE 4
JUMPING LATENCIES (SEC.) OF *Acomys cahirinus* FROM HEIGHTS OF 6.4 AND 25.4 CM TO DIFFERENT SUBSTRATES

A. All four conditions observed in one session ($N = 12$): Friedman analysis of variance

Measure	6.4 cm		25.4 cm	
	Small Checkers	Large Checkers	Small Checkers	Large Checkers
Number of Animals Jumping	12	12	2	6
Latency (includes nonjumpers)				
Median	17	12	120	118
Range	7–32	1–40	11–120	5–120
	$\chi_r^2 = 20.33$, $df = 3$, $p < 0.001$			

Sign Test Comparisons of Latencies			p (one-tailed)
Checkers	Small	6.4 cm < 25.4 cm	.003
	Large	6.4 cm < 25.4 cm	.003
Height	6.4 cm	Small = Large	ns
	25.4 cm	Small = Large	ns

B. Four sessions with one condition only in each ($n = 12$): Friedman analysis of variance

Measure	6.4 cm		25.4 cm	
	Small Checkers	Large Checkers	Small Checkers	Large Checkers
Number of Animals Jumping	8	10	3	8
Latency (includes nonjumpers)				
Median	51	23	120	80
Range	7–120	6–120	17–120	17–120
	$\chi_r^2 = 12.70$, $df = 3$, $p < 0.01$			

Sign Test Comparisons of Latencies			p (one-tailed)
Checkers	Small	6.4 cm < 25.4 cm	.09
	Large	6.4 cm < 25.4 cm	.03
Height	6.4 cm	Small > Large	.03
	25.4 cm	Small = Large	ns

cm platform to both the large and small checkered floors (Sign Test, $p = 0.003$ for both). When the stimulus combinations were presented on separate days, Group B, the latencies to jump to large from 6.4 cm were significantly shorter than those to large from 25.4 cm (Sign Test, $p = 0.03$). Although nine animals did not jump from the high platform to the small checkers, the latencies to jump from the lower platform were sufficiently long so that there was a tendency for the difference to be in the same direction as the large checkers (Sign Test, $p = 0.09$). Comparisons between Groups A and B in regard to latencies showed that the only significant difference was that Group A had shorter latencies than Group B (Median Test, $p = 0.01$) when animals were observed jumping from the low platform to the small checkers. The latencies to large checkers were significantly shorter than those to small from the 6.4-cm platform in Group B but not in Group A (Sign test, $p = 0.03$). Although more animals jumped to large checkers from 25.4 cm than to small, the latencies were not significantly different.

Exp. 4 was a partial replication of Exp. 1 and direct comparisons of the results showed three significant differences between the two experiments. For the low platform on the small checkers, the animals in Exp. 1 jumped significantly fewer times ($\chi^2 = 10.60$, $p < 0.01$) and had significantly longer latencies (Median Test, $p = 0.003$) than those of Group A from Exp. 4. As mentioned above, the animals from Group B, where the trials were spaced on separate days as in Exp. 1, also had significantly longer latencies than Group A to this stimulus combination. Spacing of the trials led the animals to be more responsive to height when trials were all on one day and more responsive to pattern when trials were separated.

The third difference was that significantly more animals jumped in Exp. 4 from the high platform ($\chi^2 = 4.36$, $p < 0.05$). This last finding may be related to the difference in ages, although within each experiment no age differences were found; to the fact that Exp. 1 animals were observed in the light phase of their lighting cycle while Exp. 4 animals were observed in the dark phase; and to the fact that the animals in Exp. 4 were born in large enclosures (100 × 100 × 90 cm) while Exp. 1 animals were born in metal cages (30.5 × 35.4 × 30.5 cm).

General Discussion

The experiments described in this paper were occasioned by the seeming discrepancy between the behavior of *Acomys cahirinus* as observed when living in their customary habitat, whether in field or laboratory, and the behavior seen in a situation structured by experimenters in the laboratory. The mice were known to jump from heights, to be able to make precise motor adjustments to distance, both vertically and horizontally, when not in the structured experimental situation. In our experiments, and that of Greenberg

(1986), *A. cahirinus* demonstrated good distance perception by jumping more frequently from a low platform than a high one; and in our experiment they jumped to a large checkered pattern more frequently than to the small checkered pattern. They also showed good distance discrimination when jumping from the 25.4-cm platform precisely to the substrate below.

That the mice in a structured experiment seemed to avoid jumping from a height that did not faze them in other situations may be related to the ambiguity of the experimental situation. The animal can only jump or remain on the platform; there is no way to escape. This may pose a stressful conflict for an animal that is in an open space that has a boundary. Thus, whether an animal jumps at all under these conditions may not be a function of depth perception alone, but of various other factors both internal and external. The failure to jump may be a pattern of adjusting to the situation for some but not for others.

During each experiment there were individuals that jumped on almost every trial and others that rarely jumped from the platform. Throughout the observations some of the animals remained immobile in the center of the platform directly above the shaft. There were also a small number of individuals that would not jump during the observation but jumped when the observer bent down to cup them for removal. These individual patterns of responding are reminiscent of the different exploratory behaviors observed by Birke and Sadler (1986). In their experiment, although there were four different stimulus configurations inside the open field in which they were observed, one-quarter of the *Acomys* left the start box slowly, returned to it frequently, and cautiously explored the arena, while all the others quickly emerged from the start box and rapidly ran around the perimeter of the arena. They concluded that these different patterns were "traits" of the individual animals because the patterns were consistent in all situations.

In our experiment there seemed to be three types of responders: those that jumped on almost every trial, those that almost never jumped, and those that would usually jump from the 6.4-cm platform but not often from the 25.4-cm platform. These differences developed over repeated trials and were not necessarily the same as the behavior exhibited on the first day when the situation was novel. We did see some rapid running around our enclosure after a jump from the platform which seems similar to that described by Birke and Sadler.

Birke and Sadler changed the stimuli inside the arena, but the arena was the same configuration leading to some environmental consistency. When given a consistent situation, whether in its living conditions or in experimental situation, animals develop consistent patterns of responding. These response patterns are specific to the entire stimulus configuration. For example, in a pilot experiment we found five young adult animals that lived with a 25.4-cm platform in their home enclosures, and frequently climbed up and jumped down from these platforms, would not jump from the same 25.4-cm platform in the experimental enclosure.

The visual cliff has been used to demonstrate depth perception in many species (Walk & Gibson, 1961). Depth or distance perception may be demonstrated not only when an animal does not jump down a great distance or jumps instead to a nearer substrate but also when an animal jumps to a distant point precisely, by making correct postural adjustments. In our experiment, most neonates and adults jumped to the shallow side of a visual cliff. In the desert, where the land is flat and open, running fast may be the first response to a predator, but to move on rocky ledges accurate depth perception and good motor agility are necessary. Our animals showed both rapid running and precise jumping. The *Acomys* in Exp. 3 had depth perception within hours of birth and the motor coordination for accurate behavioral maneuvers within two days of birth.

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